

Objective:

To find the normality of a sodium thiosulphate solution using a standard solution of potassium dichromate.

Theory:

Sodium thiosulphate solution is standardized against potassium dichromate in presence of HCl and KI. Potassium dichromate oxidizes the iodide ion in acidic medium to equivalent amount of iodine. The iodine formed in the reaction oxidizes sodium thiosulphate giving sodium tetrathionate ion and the end point is detected using starch. The iodine formed is reacted with sodium thiosulphate solution and the reaction is termed as iodimetry.



The normality of $\text{K}_2\text{Cr}_2\text{O}_7$ solution used was 0.1N

Apparatus:

1. Conical flask
2. Pipette.
3. Burette.
4. Funnel.
5. Stand and clamp.
6. Watch glass.

Chemicals used:

1. Sodium thiosulphate ($\text{Na}_2\text{S}_2\text{O}_3$)
2. Potassium dichromate ($\text{K}_2\text{Cr}_2\text{O}_7$)
3. Potassium iodide (KI)
4. Hydrochloric acid (HCl)
5. Starch solution

Procedure:

A. Preparation of $\text{K}_2\text{Cr}_2\text{O}_7$ solution:

We transfer 0.49 gm pure potassium dichromate into a 100 mL measuring flask and dissolve it

in distilled water and it is marked at a certain level. The normality of the prepared solution is measured as follows:

$$N = \frac{\text{Weight taken (in gm)} \times 0.1N}{0.49}$$

B. Finding the concentration of $\text{Na}_2\text{S}_2\text{O}_3$ solution:
We take 4 mL of 12% potassium iodide (KI) solution in a conical flask and dilute it to about 50 mL. We add about 1 gm of NaHCO_3 and shake the flask until the salt dissolves. Then, we add 4 mL concentrated HCl acid and 10 mL of standard $\text{K}_2\text{Cr}_2\text{O}_7$ solⁿ by means of a pipette in the same flask. We shake the flask and cover it with a watch glass and leave the flask in the dark for approximately five minutes. Then the watch glass is rinsed and the solution is diluted to about 100 mL. After that, the liberated iodine is titrated with sodium

thiosulphate solution from a burette until the brown colour fades to light yellow. Now, we add about 1 mL starch solution and continue the titration by adding sodium thiosulphate solution from the burette until one drop of the sodium thiosulphate solution changes the colour of the solution from deep blue to light green or light blue in colour. This is considered to be the end point of the titration. Finally, we calculate the strength / concentration of the sodium thiosulphate solution using the following equation:

$$V_1 \times S_1 = V_2 \times S_2$$

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Exp-03:
Exp Name: STANDARDIZATION OF $\text{Na}_2\text{S}_2\text{O}_3$
BY STANDARD $\text{K}_2\text{Cr}_2\text{O}_7$
SOLUTION

3/3/23

Experimental Data:

No. of obs.	Volume of $\text{K}_2\text{Cr}_2\text{O}_7$ solution taken (mL)	Burette Reading (mL)		Diff. of burette reading (mL)	Avg. volume of $\text{Na}_2\text{S}_2\text{O}_3$ solution required (mL)
		Initial	Final		
1.	10.0	0.20	10.20	10.0	9.7667
2.		10.20	19.90	9.70	
3.		19.90	29.50	9.60	

Calculation:

$$V_{\text{K}_2\text{Cr}_2\text{O}_7} \times S_{\text{K}_2\text{Cr}_2\text{O}_7} = V_{\text{Na}_2\text{S}_2\text{O}_3} \times S_{\text{Na}_2\text{S}_2\text{O}_3}$$

where, $V_{\text{K}_2\text{Cr}_2\text{O}_7}$ = Volume of $\text{K}_2\text{Cr}_2\text{O}_7$ solution = 10.0 mL

$$S_{\text{K}_2\text{Cr}_2\text{O}_7} = 0.1 \text{ N}$$

$$V_{\text{Na}_2\text{S}_2\text{O}_3} = \text{Required}$$

$$S_{\text{Na}_2\text{S}_2\text{O}_3} = \text{Concentration of } \text{Na}_2\text{S}_2\text{O}_3$$

$$V_{K_2Cr_2O_7} \times S_{K_2Cr_2O_7} = V_{Na_2S_2O_3} \times S_{Na_2S_2O_3}$$

$$\Rightarrow S_{Na_2S_2O_3} = \frac{V_{K_2Cr_2O_7} \times S_{K_2Cr_2O_7}}{V_{Na_2S_2O_3}}$$

$$= \frac{10 \times 0.1}{9.7667}$$

$$= 0.1024 \text{ N.}$$

Error Calculation:

Concentration of provided $Na_2S_2O_3 = 0.102 \text{ N}$

$$\therefore \% \text{ error} = \frac{0.1024 - 0.102}{0.102} \times 100\%$$

$$= 0.39\%$$

Results:

The practical value obtained from the experiment is,

$$S_{Na_2S_2O_3} = 0.1024 \text{ N.}$$

Discussion:

1. All equipments need to be thoroughly cleaned with distilled water before starting the experiment.
2. Starch is a polymer of glucose and it reacts to produce I_3^- (triiodide ion) which can create errors. So we must add starch at the end of the titration.
3. The reaction mixture should be kept in the dark before titration because light accelerates a side reaction in which iodide ions are oxidized to iodine by atmospheric oxygen.